ILANA SHAPIRO

CONTACT

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RESEARCH INTERESTS

My interests lie at the intersection of Programming Languages, Automated Reasoning, and AI. I am interested in developing scalable automated reasoning techniques that support trustworthy, formally verified AI systems, with a particular interest in SAT and SMT solvers.

EDUCATION

University of California, San Diego, La Jolla, CA

Ph.D. Computer Science — *Aug. 2023-present*
M.S. Computer Science — *Aug. 2023-Mar. 2026*
Programming Systems Group, advised by Professor Sorin Lerner
GPA: 3.97/4.0

Pomona College, Claremont, CA — *2018-2022*

B.A. Computer Science/Music (Flute) double major, minor in Mathematics
GPA: 4.0/4.0, Summa Cum Laude, Distinction in Senior Exercises

- ***CS Thesis:*** "MusAssist: A Domain Specific Language for Music Notation"
  - Advised at Harvey Mudd College by Professor Ben Wiedermann
- ***Music Thesis:*** "Mieczysław Weinberg: Music Transcending Tragedy"
  - Advised by Professors Alfred Cramer, Joti Rockwell, and Eric Lindholm

CONFERENCE AND JOURNAL PAPERS

Ilana Shapiro, Sorin Lerner, and Nikolaj Bjørner. "Parallelizing SMT Solvers via Dynamic Partitioning, Core-Guided Pruning, and Backbone Detection." Under review, 2026. [paper5] [code] [slides]

- At Microsoft Research, I worked with Nikolaj Bjørner on a novel parallelization algorithm for Z3. We developed an online VSIDS-based cube-and-conquer approach that uses solver threads' UNSAT cores and backbone literals for dynamic search-space pruning.

Ilana Shapiro, Ruanqianqian (Lisa) Huang, Zachary Novack, Cheng-i Wang, Hao-Wen Dong, Taylor Berg-Kirkpatrick, Shlomo Dubnov, Sorin Lerner. "Synthesizing Composite Hierarchical Structure from Symbolic Music Corpora." In *Proceedings of the 34th International Joint Conference on Artifical Intelligence (IJCAI '25),* Montreal, Canada, 2025. [paper4] [code] [slides] [talk]

- Combined stochastic and SMT techniques to frame and solve the nested NP-hard combinatorial optimization problem of music structure summarization as an extension of the Generalized Median Graph problem.

Ilana Shapiro. "MusAssist: A Domain Specific Language for Music Notation." *Proceedings of the International Conference on Technologies for Music Notation and Representation (TENOR'23),* pp. 75-82, Northeastern University, Boston, MA, 2023. [paper3] [code] [talk]

- Created external domain-specific language bridging the abstraction gap between music theoretical structures and notation. Wrote Haskell-based compiler to MusicXML.

Kaitlin Pet, **Ilana Shapiro**, and Christopher Raphael. "Virtual Ensemble Assembly: Musicality in Separation." In *Web Audio Conference (WAC'22*), Cannes, France, 2022. [paper2] [code]

- Assisted on Prof. Christopher Raphael's research at Indiana University Bloomington exploring synchronizing audio tracks without click tracks.

Ilana Shapiro and Mark Huber. "Markov Chains for Computer Music Generation." In *Journal of Humanistic Mathematics, Volume 11 Issue 2 (July 2021), pp. 167-195*. [paper1] [code]

- Created a novel system of Markov chains using inverse transform sampling, enabling end-users to rapidly generate musical sketches.

ACADEMIC THESES

Ilana Shapiro. 2022. *MusAssist: A Domain Specific Language for Music Notation*. Bachelor's thesis. Pomona College. [thesis2] [code]

- Created external domain-specific language bridging the abstraction gap between music theoretical structures and notation. Wrote Haskell-based compiler to MusicXML. [demo]

Ilana Shapiro. 2022. *Mieczysław Weinberg: Music Transcending Tragedy*. Bachelor's thesis. Pomona College. [thesis1] [recital]

- Wrote extensive musicology thesis examining narrative and memory in Weinberg's *Kaddish* Symphony. Presented flute recital of my transcriptions of Weinberg's cello works.

REPORTS AND MANUSCRIPTS

* *equal contribution*

Ilana Shapiro, Sorin Lerner, and Nikolaj Bjørner. "Z3-Parallel at the SMT Competition 2026." Technical report, 2026. [report6]

- We submitted our new Z3 parallel framework to the Parallel Track of the International Satisfiability Modulo Theories (SMT) Competition 2026.

Jarad Forristal* and **Ilana Shapiro***. "Enabling Natural Language Voice-to-Code Dictation in Talon Voice with LLMs." Unpublished manuscript, 2026. [report5] [slides]

- Enabled natural language dictation for Talon Voice, a voice-to-code technology for physically disabled programmers, via an LLM-based interpreter.

Research in Software Engineering (RiSE) Team, Microsoft Research. "Training LLMs for Verified Programming." Internship project, summer 2026. [code]

- At Microsoft Research, I helped train a 32B LLM specialized in program verification. I prepared and augmented Lean datasets for SFT and RL training, built a custom Dockerized Lean verification server, and evaluated model checkpoints.

Ilana Shapiro, Shubham Saha, Diya Lakhani, Shree Venkatesh, and Runqiu Xu. "Grid Beam Search for Constrained GPT-2 Decoding" Unpublished manuscript, 2025. [report4] [code] [slides]

- Adapted the constrained decoding algorithm Grid Beam Search (GBS) to impose lexical constraints on GPT2, and fine-tuned GPT2 on a corpus of Chekhov's stories.

Ilana Shapiro, Michael Peng, and Andrew Lara. "The Impact of GitHub Copilot on Test-First Development." Unpublished manuscript, 2024. [report3] [code]

- Conducted between-subjects pilot study to determine impact of Copilot on Test-First Development. Thematic analysis revealed that while Copilot enhanced coding speed, it resulted in superficial problem comprehension and decreased scope of the test suites.

Cole Kurashige,* Savitha Ravi,* and **Ilana Shapiro**.* "pgen-rs: LLM-Aided Efficient and User-Friendly Genomic Data Wrangling." Unpublished manuscript, 2024. [report2] [code] [slides]

- Developed pgen-rs, a tool enabling end-users to write genomic data wrangling requirements in natural language and execute with Rust-based high-performance genomic data processor.

Kyle Thompson, **Ilana Shapiro**, Ani Canumalla. "ProCon: Continuous Enumeration for Just-In-Time Bottom-Up Synthesis." Unpublished manuscript, 2024. [report1] [code]

- Introduced continuous, rule-based enumeration for just-in-time bottom-up search in SyGuS problems, where programs are enumerated in order of continuous, nonrounded weights as determined by a probabilistic weighting function.

INDUSTRY EXPERIENCE

Research Intern, Amazon Web Services (Automated Reasoning Group) *Summer 2026*

- Incoming research intern in the AWS Automated Reasoning Group working on SAT/SMT tasks.

Research Intern, Microsoft (Research in Software Engineering/RiSE Group) *Summer 2025*

- Researching SMT parallelization algorithms and natural language reasoning for formally verified code generation. Supervised by Nikolaj Bjørner.

Freelance Software Engineer, Stainless *Feb. 2023-Oct. 2024*

- Stainless automatically generates production-ready SDKs from OpenAPI specifications. I made open-source contributions to codebases including Stoplight Prism, node-tree-sitter, Microsoft Pyright, NPM Trends, and json-schema-benchmark. **Acquired by Anthropic for $300 million** in May 2026.

Software Engineer, Meta *Oct. 2022- Nov. 2022*

- Improve type safety of Python and Hack code in engineering bootcamp. Impacted by the 13% company layoff as a new hire.

3x Software Engineering Intern, Facebook *Summer 2019, 2020, 2021*

- iOS/serverside fullstack intern on Facebook Events and Groups.

TEACHING EXPERIENCE

Teaching Assistant, CSE 130: Programming Languages, UCSD (N=69) *Fall 2025*
Teaching Assistant, CSE 130: Programming Languages, UCSD (N=126) *Spring 2025*
Teaching Assistant, CS 133: Database Systems, Pomona College (N=20) *Spring 2020*

Talks

"Parallelizing SMT Solvers via Dynamic Partitioning, Core-Guided Search-Space Pruning, and Online Backbone Detection." *Programming Systems Group, UC San Diego.* May 2026

"Parallelizing Z3: Adaptive Cubing via Online Sampling of CDCL Conflict Traces." *Programming Systems Group, UC San Diego.* Aug. 2025

"Synthesizing Composite Hierarchical Structure from Symbolic Music Corpora." *The 34th International Joint Conference on Artificial Intelligence (IJCAI).* Feb. 2025

"Synthesizing Composite Hierarchical Structure from Symbolic Music Corpora." *The 19th SoCal Programming Languages and Systems Workshop (SoCaL PLS).* Nov. 2025

"Deriving Structure from Music Corpora." *Programming Systems Group, UC San Diego.* Apr. 2024

"MusAssist: A Domain Specific Language for Music Notation." *International Conference on Technologies for Music Notation and Representation.* May 2023

ACADEMIC HONORS

NSF Graduate Research Fellowship, Honorable Mention *2025*

- The NSF GRFP recognizes and supports outstanding graduate students who are pursuing full-time research-based master's and doctoral degrees in STEM fields.

The Phi Beta Kappa Award *2022*

- Endowed by the Pomona Chapter of Phi Beta Kappa, awarded to one senior selected for high quality of scholarship and promise of future distinction.

The Rena Gurley Archibald High Scholarship Prize *2022*

- Awarded to the member(s) of the Pomona College graduating class ranking highest in scholarship.

Distinction in Senior Exercise (Computer Science) *2022*

- Exceptional work on the senior exercise is awarded based on review by the entire faculty of the Computer Science Department at Pomona College.

Distinction in Senior Exercise (Music) *2022*

- Exceptional work on the senior exercise is awarded based on review by the entire faculty of the Music Department at Pomona College.

The Katherine J. Hagedorn Prize *2022*

- Awarded annually to the Pomona College student(s) demonstrating exceptional loyalty and dedication to their music studies.

Phi Beta Kappa Induction (Junior Year) - Pomona College Chapter *2021*

- 1 of 8 juniors awarded for "good moral character," distinguish in "breadth of culture," and "excellence of scholarship."

The William F. Russell Prize *2020*

- Awarded annually to the Pomona College prospective music major(s) showing substantial accomplishment and significant promise in their study of music.

SKILLS

Programming Languages: Python ● C++ ● Haskell ● TypeScript ● Java ● Objective-C ● SQL

Tools/Frameworks: LaTeX ● Git ● Functional and Object-Oriented Programming

Domain Knowledge: Programming Languages ● Automated Reasoning ● SAT/SMT Solvers ● Stochastic/Combinatorial Optimization ● Neurosymbolic Generation ● User Studies

OUTREACH AND MENTORSHIP

Mentor, UCSD Early Research Scholars Program (ERSP) *Fall 2025-Spring 2026*

- Mentored 4 undergrads on year-long research project culminating in poster session and submission to ACM SPLASH Student Research Competition.

Presenter, Harmony Hacks @ CSU San Marcos *Spring 2025*

- NSF-funded event to broaden participation of women in computing. I co-hosted a Q&A to inspire high school girls to pursue careers in CS.

Mentorship Co-Chair, GradWIC UCSD *Fall 2024-Spring 2025*

- Managed the UCSD Graduate Women in Computing Mentorship Program. Paired 174 mentees with mentors, personally mentored 2 students, hosted inclusive group activities.